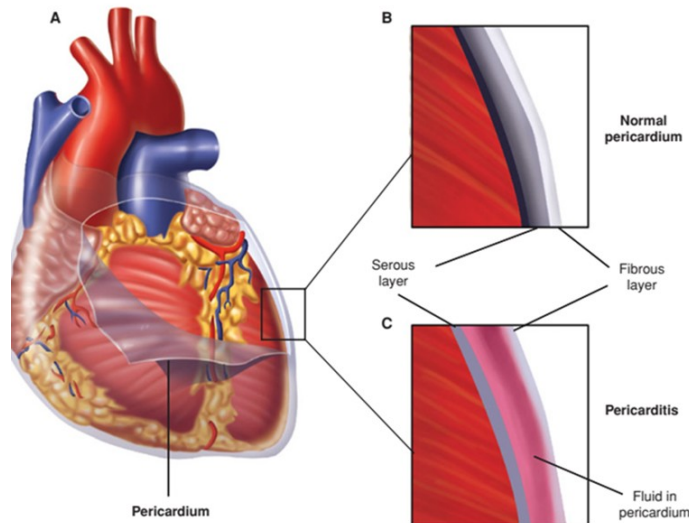
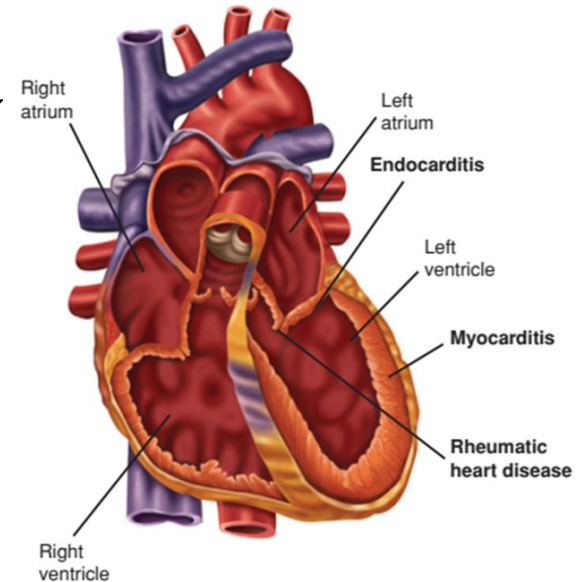


Microbial Diseases of the CVS

- Infective endocarditis (IE)
- Prosthetic valve endocarditis (PVE)
- Rheumatic heart disease (RHD)
- Pericarditis
- Myocarditis



□ Etiologic Agents

- **Native and Prosthetic valve endocarditis, RHD**

- *S. aureus*
- Coagulase-negative staphylococci
- Streptococci (+ *viridans*)
- Enterococci
- Pseudomonas
- Gram-negative rods (GNRs)
- *Coxiella burnetti*
- *Candida species*

- **Pericarditis**

- *Streptococcus pneumoniae*
- *S. aureus*
- *M. tuberculosis*
- *Coxsackie virus B*

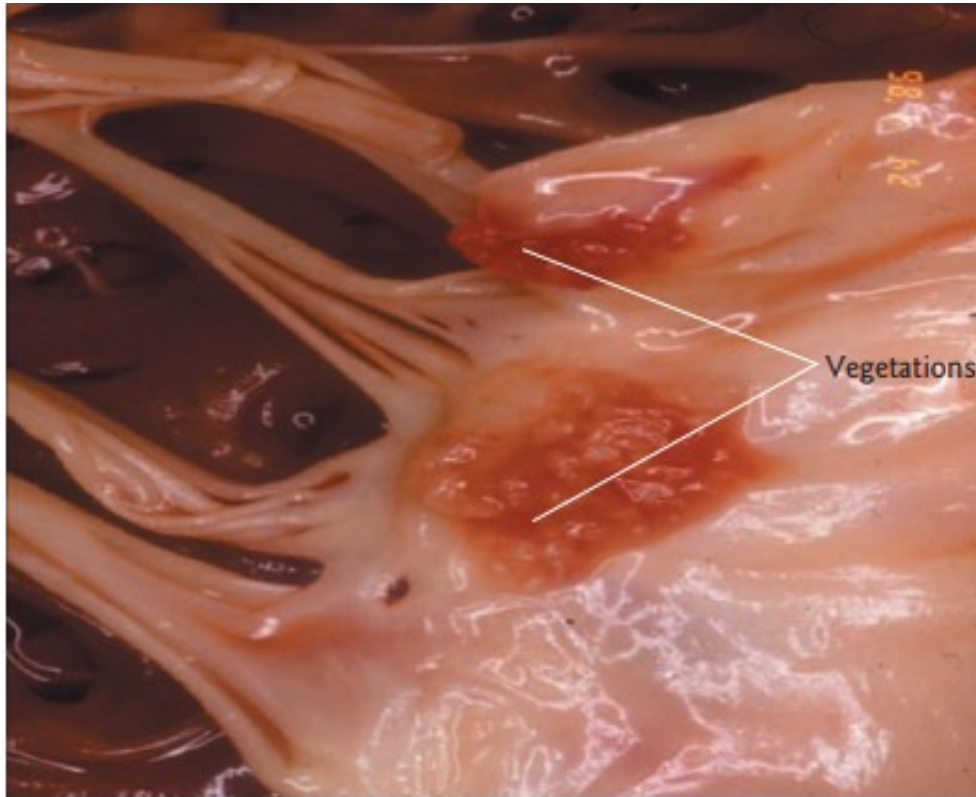
- **Myocarditis**

- *Corynebacterium diphtheriae*
- *Borrelia burgdorferi*
- *Trypanosoma cruzi*
- Coxsackie group B (less often A), enteroviruses, mumps, influenza and rubella

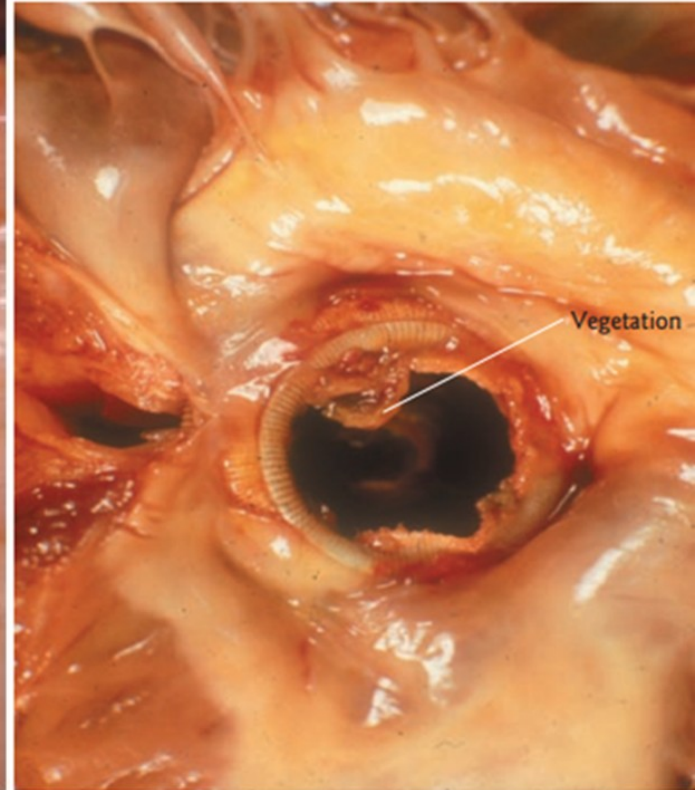
Infective Endocarditis (IE)

- an inflammation of the inner surface or endothelium of the heart.
- the surfaces most commonly affected are the surfaces of the heart valves (native or prosthetic)
- classified as infective or noninfective
 - noninfective endocarditis is very rare and can follow autoimmune reactions or can occur in patients with mucinous adenocarcinoma
- prototypic lesion of infective endocarditis is the **vegetation**
 - a mass of platelets, fibrin, microcolonies of microorganisms, and scant inflammatory cells.

Infective Endocarditis ...



Native aortic valve with bacterial vegetations



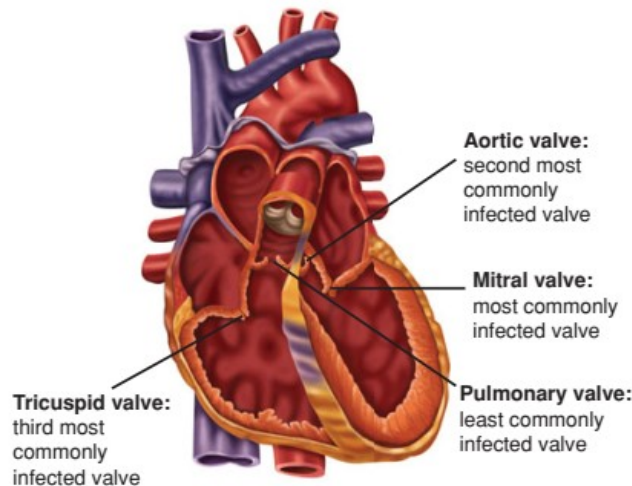
Prosthetic valve endocarditis

Infective Endocarditis ...

- IE has been classified by three major factors:
 1. **Severity and rapidity of infection**
 - acute, developing in days, (often 3–6)
 - subacute, in weeks, (often 3–6)
 - chronic, over months (often 3 or more)
 2. **Valve type**
 - native valve of the patient
 - prosthetic valve
 3. **Source of bacteraemia, if known**
 - dental, medical or surgical procedures
 - drugs: injecting drug use (IDU or IVDU)
 - devices, e.g., intravascular catheters
 - diseases elsewhere, e.g. abscesses.

Infective Endocarditis ...

- Endocarditis is further described by:
 - the infecting microorganism, usually bacterial
 - clinical type – acute, subacute or chronic
 - the infected valve(s), usually mitral or aortic, rarely tricuspid valve from IVDU
- The commonest type has been **streptococcal subacute bacterial endocarditis (SBE)** on native mitral or aortic valves



Infective Endocarditis ...

- There are six major groups of pathogens:
 - oral *Streptococci*, and *Enterococci*
 - *S. aureus*
 - coagulase-negative staphylococci (CoNS)
 - enteric Gram-negative rods (GNRs)
 - *Coxiella burnetii*
 - fungi, mainly *Candida species*

Infective Endocarditis ...

- A few bacterial species cause the majority of cases
- Pathogens vary somewhat with the clinical types of endocarditis, in part because of different portals of entry.
 - The oral cavity, skin, and upper respiratory tract are the respective primary portals for the *viridans streptococci*
 - *Staphylococci* and GNRs cause community-acquired native valve endocarditis.
 - *Streptococcus bovis* originates from the gastrointestinal tract, where it is associated with polyps and colonic tumours
 - *Enterococci* enter the bloodstream from the genitourinary tract.

Infective Endocarditis ...

- Health care-associated native valve endocarditis is the consequence of bacteraemia arising from:
 - intravascular catheter infections
 - nosocomial wound and urinary tract infections
 - chronic invasive procedures such as haemodialysis
 - Endocarditis complicates 6–25% of episodes of catheter-associated *Staphylococcus aureus* bacteraemia

Infective Endocarditis ...

Common Etiologic Agents in Infective Endocarditis

AGENT	APPROXIMATE PERCENTAGE OF CASES
Viridans streptococci (several species)	30-40
Enterococci	5-18
Other streptococci	15-25
<i>Staphylococcus aureus</i>	15-40
Coagulase-negative staphylococci	4-30
Gram-negative bacilli	2-13
Fungi (eg, <i>Candida</i> , <i>Aspergillus</i>)	2-4

Infective Endocarditis ...

Endocarditis Agents Observed in Special Circumstances

SITUATION	AGENT
Intravenous drug abuse	<i>Staphylococcus aureus</i> ; enterococci; Enterobacteriaceae and <i>Pseudomonas</i> ; fungi
Prosthetic valve infection	Coagulase-negative streptococci; <i>S aureus</i> ; Enterobacteriaceae and <i>Pseudomonas</i> ; diphtheroids; <i>Candida</i> and <i>Aspergillus</i> spp.
Immunocompromise, chronic illness	Any of the above organisms

Infective Endocarditis ...

□ Pathogenesis

- Clinical features are very varied but are basically the result of four mechanisms:
- **Valve infection**
 - altered endothelium (RHD or congenital malformations)
 - more susceptible to deposition of bacteria, platelets, and fibrin
 - a potential nidus for colonization
 - mitral or aortic valves

Infective Endocarditis ...

- **Transient bacteraemia**

- dental procedures, bronchoscopy, sigmoidoscopy, cystoscopy, and some surgical procedures
- organisms responsible are the common surface flora of the manipulated site, e.g., viridans streptococci
- *Staphylococcus aureus*, GNRs in intravenous drug abuse

Infective Endocarditis ...

• Embolization

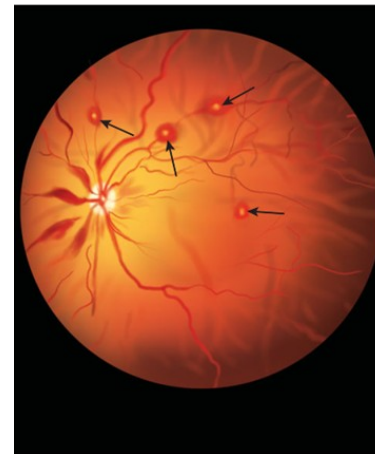
- circulating organisms adhere to the damaged surface
- complement activation, inflammation, fibrin, and platelet deposition $\Rightarrow$ further endothelial damage at the site of colonization
- entrapment of organisms leads to a mature vegetation
 - protects the organisms from host humoral and phagocytic immune defenses and from antimicrobial agents
- parts of vegetations break off and are deposited in smaller blood vessels (embolization) $\Rightarrow$ obstruction and secondary sites of infection

Infective Endocarditis ...

- **Immune complex deposition** cause peripheral manifestations
 - splinter hemorrhages under finger or toe nails, or skin or conjunctival petechiae – Roth spots
 - nephritis, cutaneous vascular lesions, arthritis



A photograph of splinter haemorrhages under finger nails



Roth spots (arrows) present on the retina of a patient with endocarditis

Infective Endocarditis ...

□ Clinical features

- Patients fall into one of five groups:
 - **Acute bacterial endocarditis**
 - fulminant with high fever and toxicity, and death may occur in a few days or weeks
 - commonly due to *S. aureus* (80%), enteric GNRs or *P. aeruginosa*.

Infective Endocarditis ...

○ Subacute bacterial endocarditis

- commonly due to *viridans streptococci*, enterococci, staphylococci
- causes weeks of fever, emboli, immunologic phenomena and cardiac murmur(s).
- progresses to death over weeks to months

○ Chronic endocarditis (rare)

- Due to fungi, Q-fever or coagulase-negative staphylococci.
- causes months of fever, splenomegaly and murmur(s).

Infective Endocarditis ...

○Prosthetic endocarditis

- occurs either
 - *early*, after operation usually due to *Staphylococci* or enteric GNRs
 - *late*, 2 months or more after operation, usually due to oral *Streptococci*, *Enterococci* or *coagulase-negative staphylococci*
- Fever, bacteraemia and a new murmur occur earlier than emboli.
- Perivalvular leaks are serious; an occluded valve is rapidly fatal.

Infective Endocarditis ...

○IVDU endocarditis

- commonly due to *S. aureus*, *streptococci*, enteric GNRs, *P. aeruginosa* or *Candida spp.*
- Fever occurs for 1–2 weeks with tricuspid valve infection and pulmonary emboli.

Infective Endocarditis ...

□ Duke Diagnostic Criteria

1. **DEFINITE** (positive blood microbiology, OR positive tissue histopathology OR 2 major criteria OR 1 major plus 3 minor, OR 5 minor criteria)
2. **POSSIBLE** (1 major plus 1 minor, OR 3 minor alone)
3. **REJECTED** (alternate diagnosis, OR response to 4 days' antibiotics, OR negative pathology)

Infective Endocarditis ...

**Table 1 Duke Endocarditis Service
Modified diagnostic criteria (abbreviated)**

Major criteria	Typical blood culture Positive echocardiogram New valvular regurgitation Positive Q fever serology
Minor criteria	Predisposition Fever Vascular phenomena Immune phenomena Suggestive microbiology/serology Positive PCR on blood or tissue (suggested) New cardiac failure or conduction defects (suggested)

Infective Endocarditis ...

- Blood cultures are essential to confirm bacteraemia
 - two *urgently* are usually sufficient in acute disease
 - three or four *in 12 hours* are usual in subacute disease
- Media must be rich with possible supplementation for fastidious organisms
- Sensitivity testing with MIC and MBC is needed as body defences in cardiac valves are minimal, so bactericidal drugs are essential

Prosthetic valve endocarditis (PVE)

- After cardiac valve prosthesis implantation, fibrin and platelet thrombi occur at the site of the sewing ring and anulus and serve as targets for adherence of microorganisms
- *Staphylococcus epidermidis* or other coagulase-negative staphylococci, *S. aureus*, diphtheroids, and fungi most often cause early-onset PVE
- This spectrum of microorganisms suggests intraoperative contamination

PVE ...

- The microbiologic cause of PVE depends on whether the infection occurs:
 - **early** (≤ 2 months) after cardiac valve replacement
 - after an **intermediate** period (2 to 12 months)
 - **late** (≥ 12 months postoperatively)
- Approximately 40% to 60% of early-onset PVE is caused by *Staphylococci*
 - *S. epidermidis* or other coagulase-negative staphylococci account for 30% to 35% of cases
 - *S. aureus* accounts for 20% to 25%
 - remaining cases are caused by gram-negative bacilli, diphtheroids, or fungi

PVE ...

- From 2-12 months, the distribution of microorganisms is similar to early-onset PVE except that:
 - the proportion of infections caused by enteric gram-negative bacilli and diphtheroids is reduced greatly
 - the frequency of streptococcal and enterococcal PVE is increased.

PVE ...

- PVE that occurs after 1 year post-op is caused by the same spectrum of microorganisms that causes community-acquired NVE except that:
 - CoNS infection occurs more commonly (~ 10% - 12% of cases) in PVE than in NVE

PVE ...

- Cases occurring within 1 year post-op most often are caused by *S. epidermidis*
 - 84% to 87% of these cases are methicillin resistant
 - at least 50% of coagulase-negative *Staphylococci* that cause PVE more than 1 year postoperatively are a species other than *S. epidermidis*, and the frequency of methicillin resistance ranges from 22% to 30%

PVE ...

- the microbial spectrum and susceptibility in vitro suggest that:
 - most cases of PVE that develop within 2 months postoperatively are nosocomially acquired
 - cases that occur from 2 to 12 months are a mix of nosocomial and community-acquired infections
 - cases that occur more than 1 year postoperatively are primarily community-acquired infections

Rheumatic fever

- follows streptococcal (*S. pyogenes*) infection, being a postinfectious complication that results from the immune response.
- streptococcal cell wall antigens stimulate antibodies that also react with cardiac sarcolemma, joint and other tissues
- acute rheumatic fever (ARF) occurs 2–4 weeks after a streptococcal throat infection
- chronic rheumatic heart disease (RHD) results from repeated attacks of *S. pyogenes* with different M antigen types
- common in developing countries

Rheumatic fever ...

- Complications of rheumatic fever include
 - heart block and cardiac failure
 - stenosis of aortic, mitral, tricuspid or pulmonary valves
 - untreated, cardiac failure, cardiomegaly and even chest deformity
 - prevented by timely prosthetic valve replacement

Rheumatic fever ...



A stenotic mitral valve from a patient with rheumatic heart disease. The valve is seen from the left atrium and shows fusion of commissures and thickening and calcification of the cusps.

Rheumatic fever ...

- children with fever and a sore throat:
 - a throat swab tested for streptococci
 - Gram stain
 - culture on sheep blood agar (small colonies, usually glistening, mucoid if encapsulated, and with hemolysis)
 - biochemical tests and Grouping
- a rising antibody titre to streptococci (ASO) is also indicative

Pericarditis

- an infection of the pericardium, a barrier that protects the heart from infections that occur in nearby tissues
- classified into:
 - purulent
 - tuberculous
 - viral
 - amoebic
 - fungal

Pericarditis ...

- Pericarditis of any type produces:
 - chest discomfort, usually with fever
 - dyspnoea from co-existent myocarditis, from cardiac tamponade (when cardiac function is impaired by pericardial fluid), or later from constriction by fibrosed or calcified pericardium
 - oedema and arrhythmias are usually absent (unlike myocarditis)

Pericarditis ...

○Purulent pericarditis

- Due to spread from pneumococcal pneumonia, staphylococci, enteric flora or other organisms after surgery or septicaemia
- Clinical features include high fever, severe chest pain and signs of sepsis
- Gram stain and culture of the pericardial aspirate define the microbial cause

Pericarditis ...

○Tuberculous pericarditis

- *M. tuberculosis*, the cause of tuberculosis, can infect the heart by local spread, usually from lung or glands
- clinically,
 - acute with little effusion
 - chronic, often with tamponade
- the etiologic pathogen is defined using AFB stain and culture

Pericarditis ...

□ Pathogenesis of Tuberculous Pericarditis

- Pericardial involvement usually develops by
 - retrograde lymphatic spread of *M. tuberculosis* from peritracheal, peribronchial, or mediastinal lymph nodes
 - hematogenous spread from primary tuberculous infection

Pericarditis ...

- The immune response to the viable acid-fast bacilli penetrating the pericardium is responsible for the morbidity associated with tuberculous pericarditis.
- Protein antigens of the bacillus induce delayed hypersensitivity responses, stimulating lymphocytes to release lymphokines that activate macrophages and influence granuloma formation.

Pericarditis ...

- The cytokine profile suggests that tuberculous pericardial effusions arise as a result of a hypersensitivity reaction orchestrated by the TH-1 lymphocytes
- Cytolysis mediated by anti-myolemmal antibodies contributes to the development of exudative tuberculous pericarditis

Pericarditis ...

○ Viral pericarditis

- the commonest form
- usually caused by **Coxsackie group B** (less often A), other **enteroviruses**, **mumps**, **influenza** or **varicella**
- Pericardial damage is the result of:
 - direct cellular damage by the infecting virus
 - destruction of virally infected cells by sensitized T lymphocytes, or antibody-dependent cell-mediated cytotoxicity
- clinically, there is mild fever and precordial discomfort, sometimes preceded by respiratory or gastrointestinal illness.

Myocarditis

- inflammation of cardiac muscle, usually caused by infection
- infectious causes are classified by the causative organism into:
 - bacterial (rare)
 - viral (commonest)
 - parasitic
- the major infections are:
 - Chagas' disease
 - Diphtheria
 - Lyme disease
 - Viral myocarditis
- in addition, rheumatic fever produces a pancarditis, including myocarditis

Myocarditis ...

○Chagas' disease

- caused by *Trypanosoma cruzi*, a flagellated protozoan transmitted by reduviid bugs
- clinically, the disease has two phases:
- **acute infection**, often in childhood causes:
 - mild fever, malaise
 - an indurated red 'chagoma' at the site of infection with regional lymphadenopathy
 - periorbital edema (Romana's sign) from conjunctival infection

Myocarditis ...

- **chronic Chagas'** disease, years or decades later, causes
 - cardiac arrhythmias
 - thromboembolism or
 - cardiac failure, often right-sided
- acute disease is diagnosed by microscopic examination of fresh blood films, while chronic disease is diagnosed serologically

Myocarditis ...

○Diphtheria

- caused by *Corynebacterium diphtheriae*
- systemic spread of bacterial toxin results in damage to the heart 1–2 weeks after infection

○Lyme disease

- caused by *Borrelia burgdorferi*, a spirochete transmitted by ticks
- an initial rash occurs at the site of infection
- disseminated disease can cause meningitis, arthritis, myocarditis
- diagnosis is clinical and serological

Myocarditis ...

○Viral myocarditis

- most commonly caused by viruses, including Coxsackie group B (less often A), other enteroviruses, mumps, influenza and congenital rubella
- can cause acute heart failure or chronic cardiomyopathy

Microbiology of Etiologic Agents

Etiologic Agents ...

- NVE, PVE and RHD
 - *S. aureus*
 - Coagulase-negative staphylococci
 - Streptococci (+ *viridans*)
 - Enterococci
 - *Coxiella burnetti*
 - Pseudomonas + Gram-negative rods (HACEK)
 - *Candida species*
- Pericarditis
 - *Streptococcus pneumoniae*
 - *S. aureus*
 - *M. tuberculosis*
 - Coxsackie virus B

Etiologic Agents ...

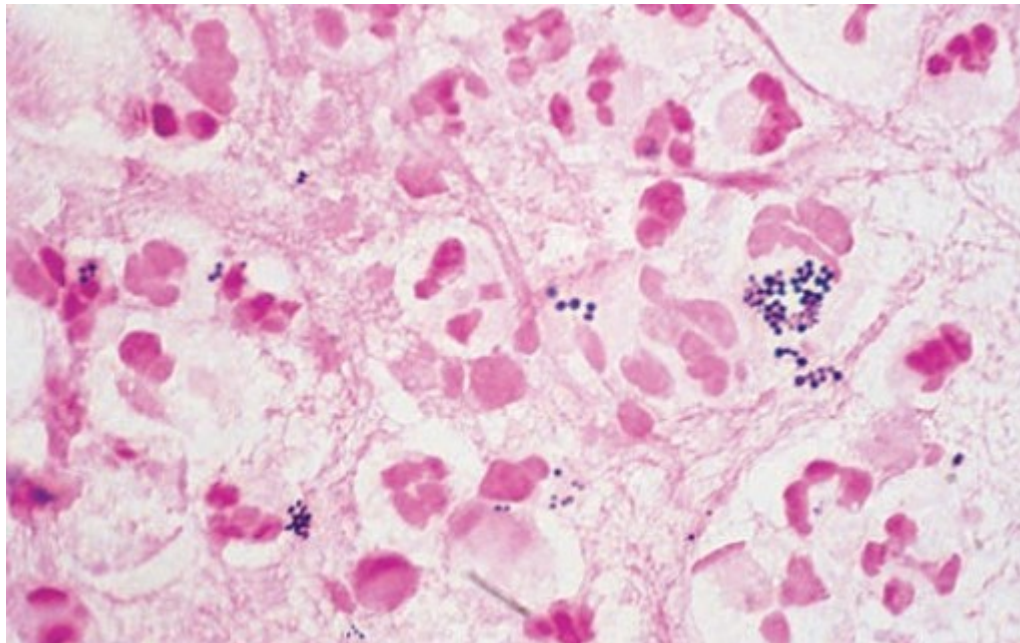
- **Myocarditis**

- *Corynebacterium diphtheriae*
- *Borrelia burgdorferi*
- *Trypanosoma cruzi*
- Coxsackie group B (less often A), enteroviruses, mumps, influenza and rubella

Gram Positive Cocci

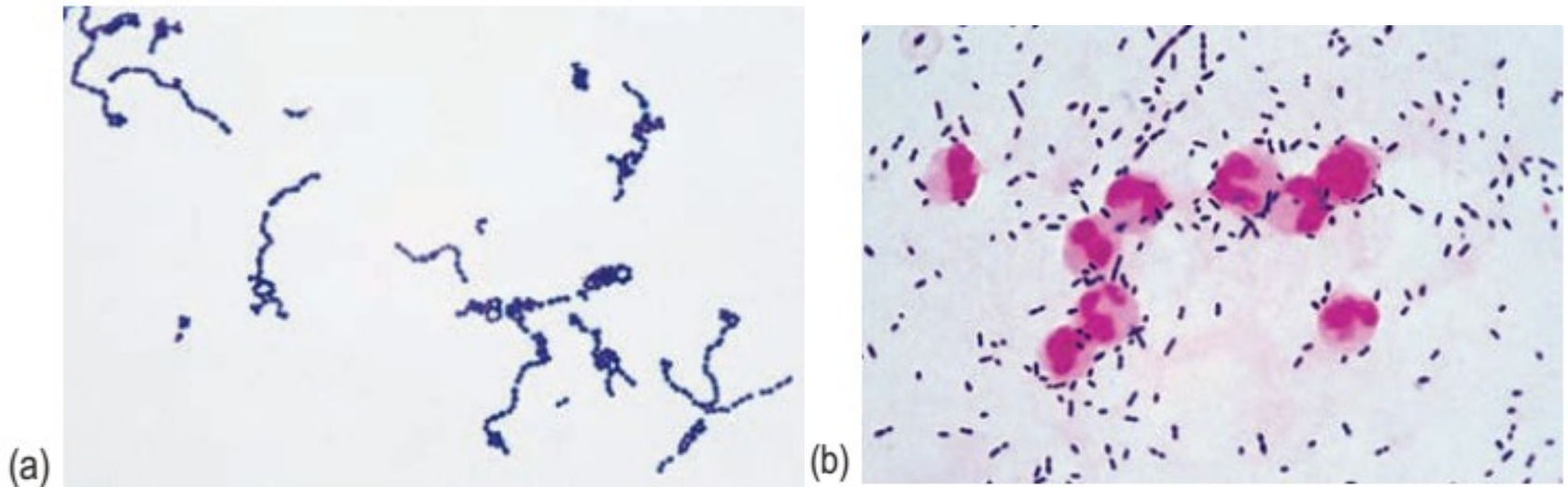
- A heterogeneous group of 21 genera
 - spherical, Gram positive, no endospores
- Based on presence or absence of catalase:
 - aerobic, catalase positive – *Staphylococcus*, *Micrococcus*, *Stomatococcus*, *Alloiococcus*
 - aerobic, catalase negative – *Streptococcus*, *Enterococcus*
 - Anaerobic - *Peptostreptococcus*

Gram Positive Cocci ...



Staphylococci singly and in small groups

Gram Positive Cocci ...



Gram stain showing *S. pyogenes* in chains (a) and *S. pneumoniae* in pairs (b)

Genus Staphylococcus

- Grow in a pattern resembling cluster of grapes in culture; but may appear singly, in pairs or short chains in clinical material
- 0.5 – 1.0 μm , non-motile, aerobic or facultative anaerobic, catalase pos., grow in media containing 10% NaCl, at 18°C-40°C
- 32 species and 15 subspecies many of which found in humans

Genus Staphylococcus...

- Important human pathogen – skin, soft tissue, bone, UT, systemic infections
- Important species: *S. aureus*, *S. epidermidis*, *S. saprophyticus*, *S. capitis*, *S. hemolyticus*
- *S. aureus* – golden colonies (carotenoid pigments); only species in humans that produces coagulase (all others coagulase-negative)

Virulence Factors

□ Structural Components

○ Capsule

- Loose-fitting polysaccharide (slime layer)
- Inhibits chemotaxis, phagocytosis by PMNs, proliferation of MNs
- Facilitates adherence to catheters and prosthetics

Virulence ...

○Peptidoglycan

- half the weight of cell wall
- layers of glycan chains built with 10-12 alternating subunits of N-AMA and N-AGA; tetrapeptide side chains attached to N-AMA subunits and cross-linked with pep. Chains
- has endotoxin-like activity: endogenous pyrogens, C' activation, IL-1 from monocytes, aggregation of PMNs (abscess formation)

Virulence ...

○Teichoic acids

- Species-specific phosphate – containing polymers bound covalently to PG
- Mediate attachment to fibronectin in mucosal surfaces
- Poor immunogens but stimulate specific Ab response
 - Dx of systemic disease by monitoring these Abs

Virulence ...

○Protein A

- Covalently linked to PG
- Affinity for Fc portions of IgG
- Prevents antibody-mediated clearance of bacterial cells
- Its detection used as specific identification test for *S. aureus*

Virulence ...

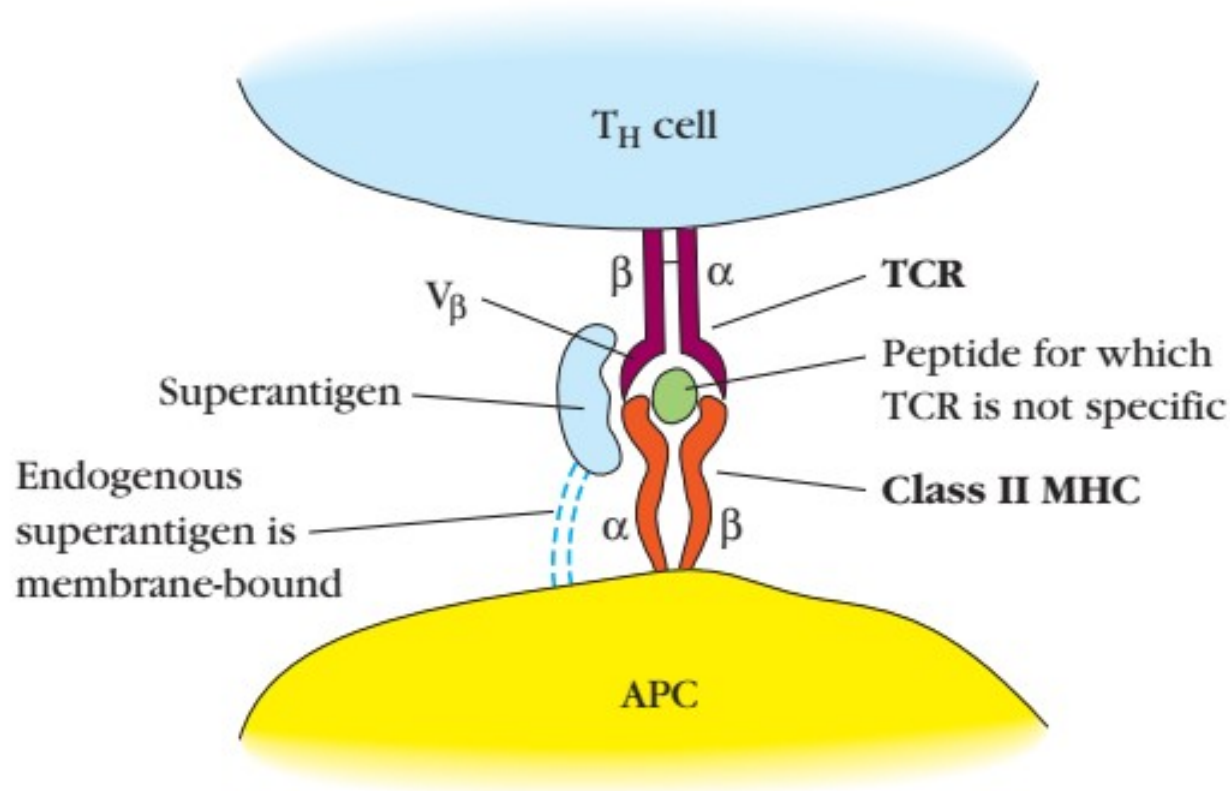
○Clumping Factor

- Bound-coagulase in *S. aureus*
- Binds fibrinogen converting it to insoluble fibrin causing staphylococci to aggregate
 - 1⁰ test for identifying *S. aureus*

Staphylococcal Toxins

- Five cytolytic (membrane-damaging): α , β , δ , γ , Panton-Valentine (P-V) toxins
- Two exfoliative toxins
- * Eight enterotoxins (A-E, G-I)
- * Toxic shock syndrome toxin – 1 (TSST-1)
- * Belong to a class of polypeptides called superantigens; bind to MHC II on APCs which interact with β -subunit of TCR $\Rightarrow$ proliferation of T-cells $\Rightarrow$ release of cytokines responsible for systemic effects

Staphylococcal Toxins...



Superantigens and the non-specific stimulation of T cell

Staphylococcal Toxins...

○ α Toxin

- Disrupts smooth muscle in blood vessels; toxic to RBCs, WBCs, hepatocytes and platelets
- Integrates to hydrophobic regions of host cell → pores → increased efflux of K^+ and influx of Na^+ and Ca^+ → osmotic lysis

○ β Toxin

- Sphingomyelinase C
- Toxic to RBCs, WBCs, macrophages, fibroblasts

Staphylococcal Toxins...

○ δ Toxin

- Cytolytic to cells and disrupts IC membrane

○ γ Toxin and P-V Leucocidin

- Bi-component toxins with two polypeptide chains
 - S (slow-eluting) – *Hl γ A*, *Hl γ C*, *LukS-P-V*
 - F (fast-eluting) – *Hl γ B*, *LukF-P-V*
- Lysis mediated by pore formation

Staphylococcal Toxins...

○Exfoliative Toxins

- 5-10% strains of *S. aureus*
- Staphylococcal Scalded Skin Syndrome (SSSS)
(also called exfoliative dermatitis, Ritter's disease)
- Two forms:
 - ETA - heat – stable, chromosomal-encoded
 - ETB - heat – labile, plasmid-encoded
- Splitting of desmosome in stratum granulosum epidermis

Staphylococcal Toxins...

○Enterotoxins

- Stable to heating at 100°C for 30 minutes
- Resistant to hydrolysis by gastric and jejunal enzymes
 - reheating contaminated food will not inactivate toxin
- 30-50% strains of *S. aureus*
- Enterotoxin-A commonly associated with disease; C and D contaminate milk products; B causes pseudomembranous colitis.

Staphylococcal Toxins...

○TSST-1

- All *S. aureus* strains responsible for menstruation-associated TSS and 50% strains responsible for other forms of TSS
- Induces non-specific cytokine release from macrophages and T-lymphocytes and increases hypersensitivity to endotoxin
- Ability to penetrate mucosal barriers in localized vaginal infections responsible for systemic effects.

Staphylococcal Enzymes

○Catalase

- *Staphylococci* produce catalase
- Converts H_2O_2 into H_2O & O_2
- Catalase test differentiates:
 - *Staphylococci* → Catalase positive
 - *Streptococci* → Catalase negative

○Coagulase

- Produced by *S. aureus*
- Clots oxalated/citrated plasma
- Deposit fibrin on the surface of Staph:
 - inhibits phagocytosis
 - associated with pathogenicity

Staphylococcal Enzymes...

○ Other enzymes

- Hyaluronidase: spreading factor
- Staphylokinase: causes fibrinolysis
- Proteinases
- β -lactamase - disrupts the β -lactam rings

Virulence Factors	Biologic Effects
<i>Structural Components</i>	
Capsule	Inhibits chemotaxis and phagocytosis
Peptidoglycan	Stimulates production of endogeneous pyrogen (endotoxin-like activity), leucocyte chemoattractant (abscess formation)
Teichoic acid	Regulates cationic conc at cell memb, binds to fibronectin
Protein A	Inhibits Ab-mediated clearance by binding to IgG, IgM, IgG ₄ FcR
<i>Toxins</i>	
Cytotoxins (α , β , δ , γ , P-V leucocidin)	Toxic for WBCs, RBCs, platelets, macrophages, fibroblasts
Exfoliative toxins (ETA, ETB)	Serine proteases that split the intercellular bridges in the epidermis
Enterotoxins (A-E, G-I)	SuperAgs (stimulate T-cell prolif and cytokine release), stimulate inflammatory mediators in mast cells, intestinal peristalsis and fluid loss
TSST-1	SuperAg (stimulates T-cell prolif and cytokine release), leakage/destruction of endoth cells
<i>Enzymes</i>	
Coagulase	Converts fibrinogen to fibrin
Catalase	Catalyses removal of hydrogen peroxide
Hyaluronidase	Hydrolyses hyaluronic acid in connective tissue, promotes spread of staph in tissue
Fibrinolysin	Dissolves fibrin clots
Lipases	Hydrolyse lipids
Nucleases	Hydrolyse DNA
Penicillinase	Hydrolyzes penicillins

Coagulase-Negative Staphylococci (CoNS)

- Comprise a large group of related species commonly found on the surface of healthy persons
- They are rarely the cause of infection
- More than 40 species are recognized
 - *S. epidermidis* accounts for about 75% of all clinical isolates
 - Other important CoNS include *S. saprophyticus* (UTI in young women)

Coagulase-Negative...

- The emergence of CoNS as major pathogens reflects:
 - increased use of implants such as intravascular lines and cannula, cardiac valves, artificial joints, etc
 - increasing numbers of severely debilitated patients in hospitals.
- CoNS are morphologically similar to *S. aureus* but
 - do not coagulate plasma
 - lack clumping factor and deoxyribonuclease.

Coagulase-Negative...

- CoNS are opportunistic pathogens that cause infection in debilitated or compromised patients such as premature neonates and oncology patients, often by colonizing biomedical devices such as intravascular lines.
- They cause particular problems in:
 - cardiac surgery (prosthetic valve endocarditis)
 - patients fitted with cerebrospinal fluid shunts (meningitis)

Coagulase-Negative...

- continuous ambulatory peritoneal dialysis (peritonitis)
- immunocompromised patients (e.g., bloodstream infection)
- intensive care units (multiple devices leading to septic shock)

Coagulase-Negative...

- Adherence to the prosthetic device by the production of an **exopolysaccharide intercellular adhesion (PIA)** is a key step in the formation of a multi-layered biofilm, essential for the pathogenesis of device-related *S. epidermidis* infection
- Important factors in the pathogenesis of *S. saprophyticus* infection include:
 - a unique adhesion protein that allows it to adhere to uroepithelial cells
 - the production of urease

Differentiation of *S. aureus* from CoNS

	<i>S. aureus</i>	<i>S. epidermidis</i>	<i>S. saprophiticus</i>
Alpha toxin (β -hemolysis)	+	-	-
Growth on 7.5% salt (mannitol salt)			
Acid from mannitol	+	-	+
Coagulase reaction	+	-	-
Pigment production (SM110)	Usually golden	Usually white	Usually white
DNase production	+	-	-
Sensitivity to novobiocin	sensitive	sensitive	resistant

Genus *Streptococcus*

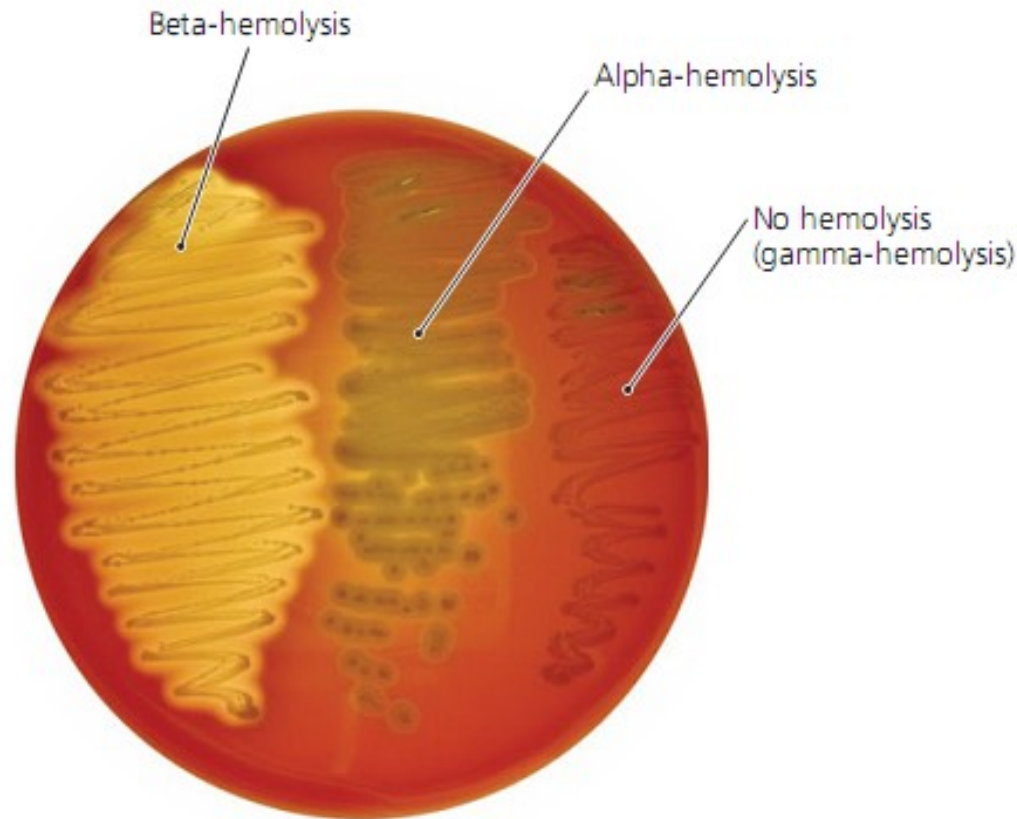
- Diverse collection of gram-positive cocci arranged in pairs or chains
- Most strains are facultative anaerobes
- Some grow only in an atmosphere enhanced with carbon dioxide
- Have complex nutritional requirements – blood- or serum-enriched media
- Catalase negative

Genus *Streptococcus*...

- Important human pathogens
- Differentiation of species within genus complicated
- Three different schemes used for classification:
 - **Serologic (Lancefield) groupings** based on group-specific carbohydrate in cell wall: A to H, K to M, and O to V.
 - **Hemolytic patterns** on blood agar: complete (β) hemolysis, incomplete (α) hemolysis, and no (γ) hemolysis
 - **Biochemical (physiologic)** properties

Classification of Common Streptococcal Pathogens

Biochemical Classification	Serologic Classification	Hemolysis Patterns
<i>S. pyogenes</i>	A	Beta (complete)
<i>S. anginosus</i> group	A, C, F, G, NG	Beta; occasionally α or non
<i>S. agalactiae</i>	B	Beta; occasionally α or non
<i>S. dysgalactiae</i>	C, G	Beta
<i>S. bovis</i>	D	α ; non; occasionally beta
Viridans group	NG	α or non
<i>S. pneumoniae</i>	NG	α



Streptococcus pyogenes (left) completely uses red blood cells, producing a clear zone termed beta-hemolysis.

Streptococcus pneumoniae (middle) partially uses red blood cells, producing a discoloration termed alpha-hemolysis.

Enterococcus faecalis (right) does not use red blood cells; the lack of any change in the medium around colonies is termed gamma-hemolysis, even though no red cells are hemolyzed

Group A beta -Hemolytic Streptococci (*Streptococcus pyogenes*)

- The most clinically important member
- The most frequently encountered bacterial pathogen of humans worldwide
- It can invade apparently intact skin or mucous membranes, causing some of the most rapidly progressive infections known
- A low inoculum suffices for infection
- Some strains cause post-infectious sequel, including rheumatic fever and acute glomerulonephritis.

Group A beta...

- Nasopharyngeal carriage is common especially in colder months and particularly among children.
- Its habitat is infected patients, and also normal human carriers in whom the organism resides on skin and mucous membranes.
- Person-to-person spread by respiratory droplets or through breaks in skin

Group A beta-Hemolytic...

☐ Physiology and Structure

- Isolates of *S. pyogenes* are 0.5 – 1.0 μm , spherical cocci that form short chains in clinical specimens and longer chains when grown in liquid media
- Optimal growth on enriched blood agar media
- 1 – 2 mm white colonies with large zones of β -hemolysis after 24 hours of incubation
- Encapsulated strains appear mucoid
- Non-encapsulated colonies are smaller and glossy

Group A beta-Hemolytic...

□ Group - specific carbohydrate

- Contained in the cell wall of many *Streptococci*
- Forms basis of serologic grouping (Lancefield groups A-U)
- Serologic specificity of the group specific carbohydrate is determined by amino sugar

Virulence factors

□ M - protein

- Major virulence factor for gp A *S. pyogenes*
- Gp A *Streptococci* that lack M- protein are not virulent
- > 80 types of M-proteins

□ Nucleoproteins

- Make up most of streptococcal cell body

Virulence factors...

☐ Toxins and enzymes

○ Streptokinase (Fibrinolysin)

- digests fibrin and other proteins

○ Deoxyribonuclease (DNase)

- depolymerizes DNA

○ Hyaluronidase

- splits hyaluronic acid; important component of connective tissue
- serves as spreading factor

Virulence factors...

○Pyrogenic exotoxins A - C (Erythrogenic toxin)

- heat-labile
- superantigens that interact with macrophages and TH-cells → IL-1, IL-2, IL-6, TNF- α , TNF- β , IFN- γ → rash in scarlet fever and shock and organ failure in toxic shock syndrome

○Diphosphopyridine Nucleotidase

- seen in some streptococci
- related to organism's ability to kill leukocytes

Virulence ...

○ Hemolysins

A. Streptolysin O

- oxygen-labile hemolysin
- lyses RBCs, platelets, leucocytes
- antibodies formed against it (anti-streptolysin O)

B. Streptolysin S

- oxygen-stable, non-immunogenic, cell-bound hemolysin that lyses RBCs, platelets, leucocytes
- responsible for the characteristic β -hemolysis on blood agar

Virulence Factors of *S. pyogenes*

Virulence Factor	Biologic Effect
Capsule	Antiphagocytic
LTA	Binds to epithelial cells
M protein	Adhesin; antiphagocytic; degrades C3b
M-like protein	Binds IgG, IgM and α -2-macroglobulin
F protein	Mediates adherence to epithelial cells
Pyrogenic exotoxins	Mediate pyrogenicity; enhance DTH; enhance cytotoxicity; mitogenic to T cells
Streptolysin S	Lyses leucocytes, platelets, RBCs; release of lysosomal enzymes
Streptolysin O	Same effect as Streptolysin S
Streptokinase	Lyses clots; facilitate bacterial spread
DNase	Depolymerizes cell-free DNA in pus

Laboratory Diagnosis

- Rapid latex antigen kits for direct detection of group A streptococci in patient samples are widely used.
- Specimens from patients with clinical signs of pharyngitis and a negative antigen detection test should undergo routine culturing for streptococcal identification.
- Depending on the form of the disease, specimens for laboratory analysis can be obtained from throat swabs, pus and lesion samples, sputum, blood, or spinal fluid.

Laboratory...

- *S. pyogenes* forms characteristic small, opalescent colonies surrounded by a large zone of beta hemolysis on sheep blood agar.
- Bacitracin susceptibility test (sensitive)
- L-pyrrolidonyl arylamidase (PYR) test
 - * Pyrrolidonyl aminopeptidase, is a bacterial enzyme that hydrolyzes L-pyroglutamic acid- β -naphthylamide impregnated into the test disk that serves as a substrate. Hydrolysis of the substrate yields beta-naphthylamine which combines with the PYR Reagent (p-dimethylamino-cinnamaldehyde) to form a bright pink to cherry red color.
- Serologic tests detect a patient's antibody titer to streptolysin-O (ASO test)

Group B beta-Hemolytic Streptococci (*Streptococcus agalactiae*)

- Found in the vaginal tract of female carriers, and the urethral mucous membranes of male carriers, as well as in the GI tract.
- Transmission occurs from an infected mother to her infant at birth, and venereal (propagated by sexual contact) among adults.
- Leading cause of meningitis and septicemia in neonates, with a high mortality rate.

Group B beta...

- Samples of blood, cervical swabs, sputum, or spinal fluid can be obtained for culture on blood agar.
- Group B streptococci are β hemolytic, with larger colonies and less hemolysis than group A.
- Identification by positive CAMP test and hippurate hydrolysis
 - GBS produce CAMP factor that enhances β hemolysis of *S. aureus* when a streak of the later is exposed to CAMP factor

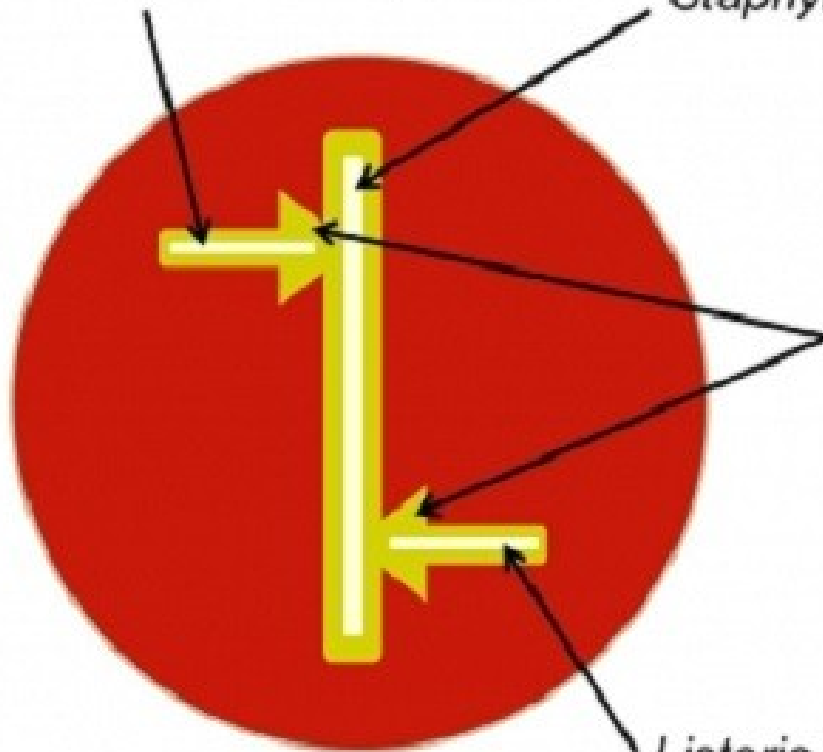
Group B beta...

- CAMP test detects the production of diffusible, thermostable, extracellular protein known as CAMP factor, produced by *Group B Streptococcus*.
- The CAMP factor acts synergistically with the beta lysin produced by *Staphylococcus aureus* to produce a zone of enhanced lysis of sheep or bovine erythrocytes.
- CAMP test depends on the elaboration of two toxins during growth to form a typical arrow-head or flame-shaped clearing at the junction of the two organisms when they are placed perpendicular to each other.

Group B beta...

Group B *Streptococcus* spp.

Staphylococcus aureus (β -hemolysis)



Increased zone of β -hemolysis due to CAMP factor production

Listeria monocytogenes

Non-groupable α -hemolytic *Streptococci*

□ *Streptococcus Pneumoniae* (Pneumococcus)

- *S. pneumoniae* are gram-positive, non-motile, encapsulated cocci.
- They are lancet-shaped, and their tendency to occur in pairs accounts for their earlier designation as *Diplococcus pneumoniae*.
- They are surrounded by a polysaccharide capsule.
- *S. pneumoniae* is the most common cause of pneumonia and otitis media and an important cause of meningitis and **bacteremia**

S. pneumoniae...

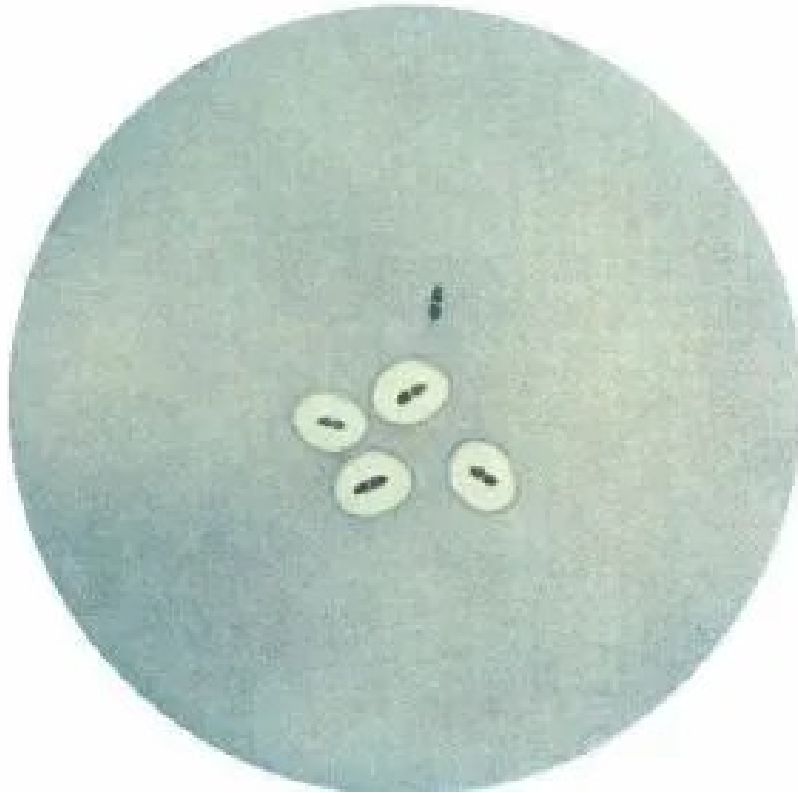
❑ Laboratory Tests

- Specimens for laboratory evaluation can be obtained from a nasopharyngeal swab, blood, pus, sputum, or spinal fluid.
- Growth is facilitated in 5-10% CO₂ atmosphere.
- α Hemolytic colonies appear when *S. pneumoniae* is grown on blood agar overnight under aerobic conditions at 37°C.

S. pneumoniae...

- Lancet-shaped, gram-positive diplococci are observed on a Gram stain of the sample.
- Growth of these bacteria is inhibited by low concentrations of the optochin, and the cells are lysed by bile acids (bile solubility positive).
- Capsular swelling is observed when the pneumococci are treated with type-specific antisera (Quellung reaction).

S. pneumoniae...



Quellung reaction: capsule appears as a sharply demarcated halo around the dark blue stained cell

Non-groupable α -hemolytic ...

□ *Viridans Streptococci*

- Normal inhabitants of oral cavity, GIT and female genital tract
- Include *S. sanguis*, *S. salivarius*, *S. mitis*, *S. intermedius*
- Associated with:
 - Subacute bacterial endocarditis in patients with prosthetic valves
 - Deep-seated abscesses in liver and brain
 - Female genital infections
 - Oral abscesses
 - Endocarditis, ARDS and shock in neutropenic patients

Genus Enterococcus

- Gram-positive cocci that occur in singles, pairs, and short chains
- Contain Lancefield group D Ag and were considered group D streptococci in the past



Enterococcus...

- Sufficient genetic evidence to show their difference from members of genus *Streptococcus*
- Facultative anaerobes
- Able to grow under rather extreme conditions - in 6.5% NaCl at pH 9.6; at temperatures ranging from 10° C to 45° C.; in the presence of 40% bile salts

Genus Enterococcus...

- Most clinical isolates of *Enterococci* are:
 - *E. faecalis*, which accounts for 80% to 90%
 - *Enterococcus faecium* accounts for 5% to 10% of isolates.
- *Enterococci* are not very virulent, but they have become prominent as causes of nosocomial infections as a result of their multiple antibiotic resistance.
- Part of the normal faecal flora, however, they can also colonize oral mucous membranes and skin, especially in hospital settings.
- Are highly resistant to environmental and chemical agents, and can persist on fomites.

Genus Enterococcus...

- Elaborate a number of virulence factors that contribute to the infection process:
 - *E. faecalis* elaborates **cytolysin, gelatinase, aggregation substance, and an enterococcal surface protein (Esp)**
 - enterococcal cytolysin is a proinflammatory, acutely lethal haemolytic toxin, which also functions as a bacteriocin against other Gram-positive bacteria.
 - the aggregation substance is a surface protein that
 - promotes aggregate formation during bacterial conjugation
 - is associated with binding to, and subsequent intracellular survival within, human neutrophils

Genus Enterococcus...

○ Diseases

- Seldom cause disease in normal, healthy individuals.
- Under conditions in which host resistance is lowered or the integrity of the gastrointestinal or genitourinary tract has been disrupted (e.g., by instrumentation),
- *Enterococci* can spread to normally sterile sites, causing:
 - urinary tract infections
 - meningitis
 - bacteriaemia
 - neonatal sepsis
 - Sub-acute bacterial endocarditis
 - biliary tract infection
 - intra-abdominal abscesses

Genus Enterococcus...

○Laboratory identification

- Ability to survive in the presence of bile
- Ability to hydrolyze the polysaccharide esculin
- Ability to grow in 6.5 % NaCl
- Positive Pyrrolidonyl Aminopeptidase (PYR) test - hydrolysis of L-pyroglutamic acid- β -naphthylamide to form a bright pink to cherry red colour

Pseudomonas aeruginosa

- *P. aeruginosa* is widely distributed in nature and is commonly present in moist environments in hospitals
- It can colonize normal humans, in whom it is a saprophyte
- *P. aeruginosa* is motile and rod-shaped, measuring about $0.6 \times 2 \mu\text{m}$.
- It is gram-negative and occurs as single bacteria, in pairs, and occasionally in short chains

Pseudomonas aeruginosa...

- *P. aeruginosa* is an obligate aerobe that grows readily on many types of culture media, sometimes producing a sweet or grape-like or corn taco-like odor.
- Some strains hemolyze blood
- *P. aeruginosa* forms smooth round colonies with a fluorescent greenish color
- It often produces the non-fluorescent bluish pigment **pyocyanin**, which diffuses into the agar

Pseudomonas aeruginosa...

- Many strains of also produce the fluorescent pigment **pyoverdinin**, which gives a greenish color to the agar
- Some strains produce the dark red pigment **pyorubin** or the black pigment **pyomelanin**
- *P. aeruginosa* grows well at 37–42 °C; is oxidase-positive; does not ferment carbohydrates
 - identification is usually based on colonial morphology, oxidase positivity, the presence of characteristic pigments, and growth at 42 °C

Pseudomonas aeruginosa...

□ Virulence Factors

- Low intrinsic virulence for humans
- A wide array of cell-associated and secreted factors:
 - Adhesins
 - mediate the attachment of *P. aeruginosa* to host tissues and mucins
 - four groups: pili, mucin-binding outer membrane (OM) protein F, surface lectins and mucoid exopolysaccharide (alginate)

Pseudomonas aeruginosa...

○ Flagella

- promote chemotaxis and motility during invasion of the tissues
- function as ligands for macrophages and other phagocytic cells
- tether the organism to epithelial membranes

Pseudomonas aeruginosa...

○Lipopolysaccharide (endotoxin)

- plays a major role in protecting the cell from the complement-mediated bactericidal action of normal human serum
- composed of an external unit membrane (OM), a layer of peptidoglycan and an inner cytoplasmic membrane

○Exopolysaccharide/Alginate

- an extracellular slime that aggregates loosely around the cell
- responsible for the mucoid colonies seen in cultures from patients with CF

Pseudomonas aeruginosa...

○ Exoenzymes

• Exotoxin A (ETA)

- produced by 90% of the species
- closely related to diphtheria toxin
- catalyzes the transfer of the ADP-ribosyl moiety of oxidized NAD (NAD^+) to elongation factor 2 (EF2). This reaction inactivates EF2, terminates peptide chain elongation and leads to the inhibition of protein synthesis and cell death
- composed of two fragments: Fragment A has enzymatic activity; Fragment B acts as the binding moiety

Pseudomonas aeruginosa...

- **Exoenzyme S**

- catalyzes the transfer of ADP-ribose from NAD⁺ to other eukaryotic proteins
- more resistant to heat
- it is not neutralized by antibodies to ETA

Pseudomonas aeruginosa...

○Hemolysins

- two types: heat-labile phospholipase C (PLC) and a heat-stable rhamnolipid
- PLC degrades phospholipids containing quaternary ammonium groups such as phosphatidylcholine, the major phospholipid component of lung surfactant

○Proteolytic enzymes

- active against a variety of substrates such as gelatin, casein, elastin, collagen and fibrin
- three classes of protease have been identified: a general protease, elastase and alkaline protease

Pseudomonas aeruginosa...

○Cytotoxin

- localized to the periplasmic space as an inactive form
- activated by proteases
- linked to the observed lack of leucocytes during sepsis due to *P. aeruginosa* and the fall in the granulocyte count

Pseudomonas aeruginosa...

○Pyochelin and Pyoverdine

- iron chelators
- play a key role in the growth and virulence of the species in iron-limited conditions

○Pyocyanin

- highly toxic for human respiratory epithelial cilia
- induces ciliary dyskinesia and disrupts the integrity of the epithelial surface

Pseudomonas aeruginosa...

□ Pathogenesis and Clinical Manifestations

- *P. aeruginosa* is pathogenic only when introduced into areas devoid of normal defenses:
 - disruption of mucous membranes and skin
 - use of intravenous or urinary catheters
 - neutropenia as in cancer chemotherapy
- the bacterium attaches to and colonizes the mucous membranes or skin, invades locally, and produces systemic disease
 - promoted by the pili, enzymes, and toxins

Pseudomonas aeruginosa...

- LPS plays a direct role in causing fever, shock, oliguria, leukocytosis and leukopenia, DIC, and adult respiratory distress syndrome
- *P. aeruginosa* produces:
 - infection of wounds and burns, giving rise to blue-green pus
 - meningitis, when introduced by lumbar puncture
 - urinary tract infection, when introduced by catheters and instruments or in irrigating solutions
 - necrotizing pneumonia from contaminated respirators

Pseudomonas aeruginosa...

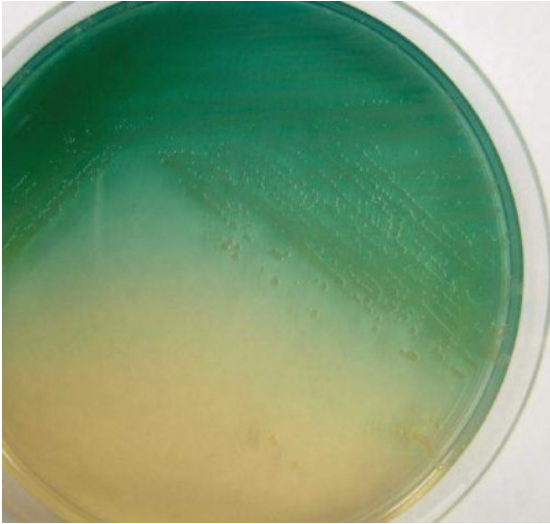
- mild otitis externa in swimmers
- invasive (malignant) otitis externa in diabetic patients
- infection of the eye, which may lead to rapid destruction of the eye, occurs most commonly after injury or surgical procedures
- fatal sepsis in infants, in patients with leukemia or lymphoma who have received antineoplastic drugs or radiation
- hemorrhagic necrosis of skin called **ecthyma gangrenosum**

Pseudomonas aeruginosa...

□ Laboratory Diagnosis

- **Specimens:** pus, urine, sputum, blood, eye swabs, surface swabs.
- **Gram Stain:** Gram-negative rods.
- **Culture**
 - Obligate aerobe, grows readily on all routine media over wide range of temperature (37-42°C)
 - Bluish-green pigmented large colonies with characteristic “fruity” odour on culture media
- **Biochemical:** *P. aeruginosa* does not ferment lactose

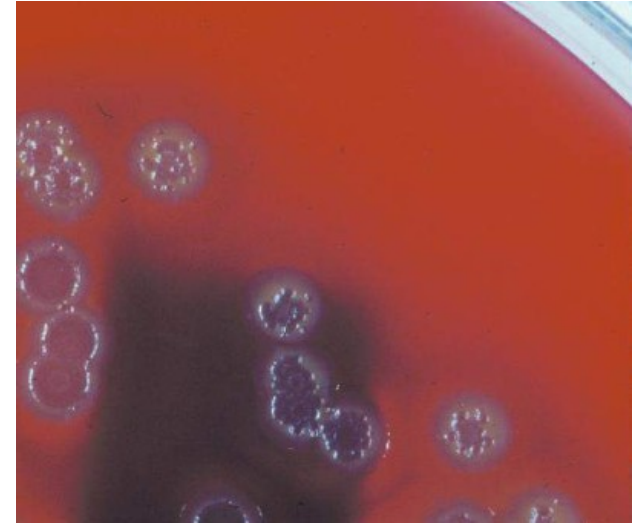
Pseudomonas aeruginosa...



The organism produces pyocyanin, which is blue, and pyoverdinin, which is green



Green-grey colonies 6–8 mm in diameter on blood agar

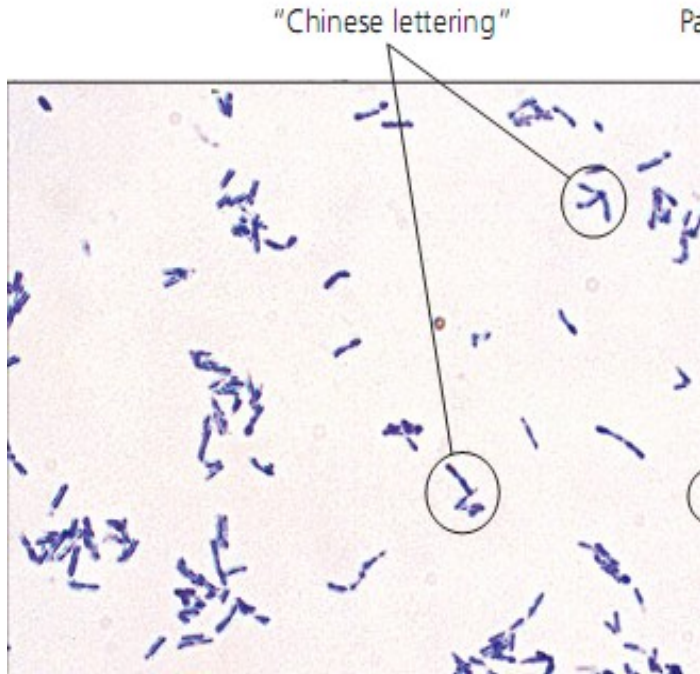


Silver-toned dry colonies on a similar blood agar plate; no hemolysis is present

Corynebacterium diphtheriae

- Gram – positive, non-capsulated, non-motile, and non-spore forming pleomorphic bacilli
- Characteristically, they possess irregular swellings at one end that gives them “club-shaped” appearance
- Irregularly distributed within the rod (often near the poles) are granules staining deeply with aniline dyes (metachromatic granules) that give the rod a beaded appearance

Corynebacterium diphtheriae...



- Individual *Corynebacterium* in stained smears tend to lie parallel or at acute angles to one another resembling Chinese letters

Corynebacterium diphtheriae...

○ *C. diphtheriae* biovars

- four biovars (biotypes): *gravis*, *intermedius*, *mitis*, and *belfanti*.
 - originally these names were used to describe the severity of disease.
- toxigenic and non-toxigenic strains occur in all *C. diphtheriae* biovars.

Virulence Factors and Pathogenesis

- major virulence of *C. diphtheriae* results from the action of its potent exotoxin (**diphtheria exotoxin**), which inhibits protein synthesis in mammalian cells
- the polypeptide toxin is comprised of two segments:
 - B, which binds to specific receptors on susceptible cells
 - A, the active segment
- Following proteolytic cleavage of the bound molecule, segment A enters the cell, where it catalyses inactivation of the transfer RNA (tRNA) translocase

Virulence factors...

- the toxin affects all cells in the body, but the most prominent effects are on the heart (**myocarditis**), nerves (demyelination), and kidneys (tubular necrosis)
- a second toxin, **dermonecrotic toxin**, is a sphingomyelinase acting on vascular endothelial cells, increasing vascular permeability

Coxiella burnetii

- Gram-negative rod
- Grows intracellularly in eukaryotic cells
- Two structural forms:
 - Small cell variants (Phase I)
 - resistant to env'tal stress
 - have intact LPS
 - Large cell variants (Phase II)
 - have a LPS with lipid A and core sugars but lacks O-Ag sugars

Coxiella ...

- Associated with arthropods (ticks)
- Primary reservoirs for human disease are farm animals and pets (cats and dogs)
- Human infection occurs after:
 - inhalation of airborne particles from contaminated environment
 - ingestion of contaminated unpasteurized milk and dairy products
- Ticks do not transmit disease to humans

Coxiella ...

- Pathogenicity related with ability to undergo phase transition
 - Phase I variants arrest fusion of phagosomes with endosomes and then with lysosomes $\Rightarrow$ survival
- Clinical diseases
 - asymptomatic infection in majority
 - mild flu-like symptoms (acute Q fever) – abrupt onset, HGF, fatigue, headache, myalgia
 - pneumonia and hepatitis in 5%
 - chronic Q fever (months or years after initial exposure)
 - **subacute endocarditis** in underlying valvular disease

Coxiella ...

- **Diagnosis**
 - Tissue culture (rarely)
 - Serology (MAT, IFA, ELISA) to demonstrate antibodies to Phase I and/or Phase II antigens
 - IgM and IgG antibodies against Phase II in acute Q fever
 - Antibodies to both Phase I and II with higher titers to Phase II in chronic Q fever
 - PCR

Candida Species

- *Candida* are small (4–6 μm), oval, thin-walled yeast-like fungi that reproduce by budding or fission
- The genus *Candida* is comprised of over 200 species and constitutes an extremely diverse yeast species whose common bond is the absence of a sexual cycle
- Only a few species cause disease in humans
- The medically significant *Candida* species include: *Candida albicans*, *Candida glabrata*, *Candida parapsilosis*, *Candida tropicalis*, *Candida krusei*, *Candida kefyr*, *Candida guilliermondii*, *Candida lusitanae*, *C. stellatoidea*, and *Candida dubliniensis*

Candida Species ...

○ Virulence factors

- *Candida albicans* undergoes reversible morphological transition between budding pseudohyphal and hyphal growth forms
 - yeast cells may be disseminated more effectively
 - hyphae promote invasion of epithelial and endothelial tissue and help evade macrophage engulfment
- adhesive ability of *C. albicans* has been correlated with pathogenesis of infection
 - cell-surface adhesion glycoproteins

Candida Species ...

- invasion of host cells by *Candida* involves penetration and damage of the outer cell envelope
 - phospholipases, by cleaving phospholipids, induce cell lysis and thereby facilitate tissue invasion
- ability of *Candida* species to overcome the suppressive effect of antifungal chemotherapy is considered a virulence attribute

Candida Species ...

○Pathogenesis

- *Candida myocarditis* is the result of hematogenous dissemination with development of one or more abscesses within the myocardium
- *Candida spp.* may reach the pericardium
 - from adjacent endocarditis or myocarditis
 - via hematogenous seeding or direct inoculation at the time of cardiac surgery

Candida Species ...

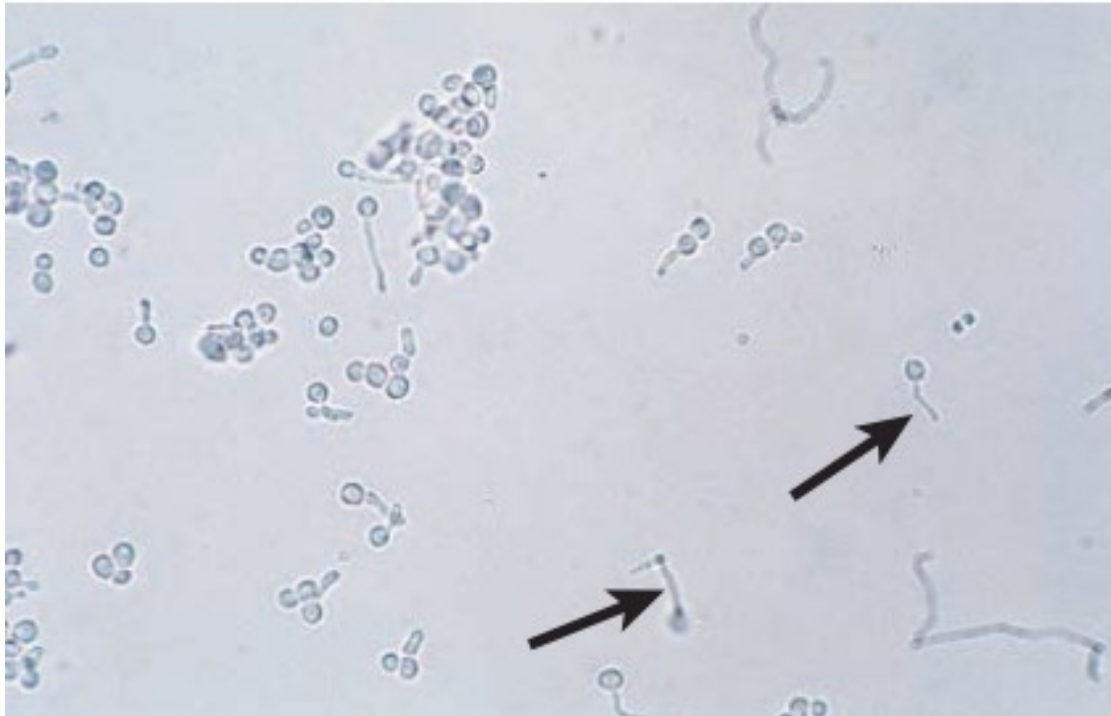
- *Candida* organisms rarely adhere to and colonize normal valvular endothelium to cause endocarditis
 - predisposing factors include:
 - underlying valvular disease
 - prosthetic cardiac valves, including homograft, which provide an abnormal surface for *Candida* adherence once coated with platelets and fibronectin
 - prolonged presence of intravascular catheters, resulting in persistent high inoculum candidemia
 - heroin addiction (R-side endocarditis)

Candida Species ...

○Lab. Diagnosis

- Typical Gram stain, culture on usual laboratory media such as blood agar
- Germ tube formation in serum in 2–3 hours
- Carbohydrate assimilation and fermentation tests for other species
- Serologic assays include testing for *Candida* antibodies, antigens and metabolites - *Candida* mannan antigen, *Candida* heat labile glycoprotein antigen, *Candida* enolase antigen, D-arabinatol

Candida Species ...



Group B Coxsackie Virus

- a member of the family Picornaviridae, genus Enterovirus
- non enveloped, positive-sensed, single-stranded RNA virus
- responsible for several syndromes:
 - Exanthema
 - Aseptic meningitis
 - Encephalitis
 - Pleurodynia
 - Myocarditis and pericarditis
 - Pancreatitis leading to beta-cell destruction and type one diabetes mellitus

Group B Coxsackie ...

- The pathogenesis of coxsackieviruses depends upon specific virus-receptor interactions
 - determine the location and the origin of the primary infection
 - influence the viral spread to other organs during the post-viremic stage
- **The coxsackievirus-adenovirus receptor (CAR) and the decay-accelerating factor (DAF) are the notable receptor proteins that play an important role in the pathogenesis of Coxsackie B virus infections**

Group B Coxsackie ...

- DAF is expressed in epithelial and endothelial cells
- CAR is present in intercalated discs
 - structures that link the myocardial cells and relay signals between cells
 - interactions with these receptors facilitate Coxsackie B virus entry into the myocardial cells, eventually leading to myocarditis
- Diagnosis is mainly by RT-PCR, viral cultures, or serological tests